

Topic 3 Review [98 marks]

1. [Maximum mark: 3] 25M.1.SL.TZ2.2
Explain **one** reason why a virtual private network (VPN) might be used. [3]

2. [Maximum mark: 15] 25M.1.SL.TZ2.10
A school has a local area network (LAN) with a central server that stores many files containing personal, health, and financial information.

The LAN is used by the following types of user: network administrators, teachers, students and guests.

- (a) Explain how the different levels of access for the users of this LAN could be implemented. [6]
- (b.i) Suggest **one** communications link that would provide high-speed internet access for the school. [2]
- (b.ii) Suggest **two** measures to protect the school's LAN from external network security threats. [4]

The school gives students and teachers their own school email account.

- (c) List **three** problems that might result from providing email access to all students and teachers. [3]
3. [Maximum mark: 2] 24N.1.SL.TZ0.5
Outline what is meant by a media access control (MAC) address. [2]
4. [Maximum mark: 2] 24M.1.SL.TZ1.12

Identify **two** of the layers of the Open Systems Interconnection (OSI) seven-layer model.

[2]

5. [Maximum mark: 15]

24M.1.SL.TZ1.13

Different transmission media may be used within a network.

(a.i) Identify **two** characteristics of fibre optic cables as a transmission medium.

[2]

(a.ii) Identify **two** characteristics of wireless transmission.

[2]

(b) Describe how encryption is used to protect data during transmission.

[3]

(c) Explain how data is transmitted using packet switching.

[5]

(d) Explain **one** social implication of changes to working patterns due to the use of a virtual private network (VPN).

[3]

6. [Maximum mark: 3]

23N.1.SL.TZ0.4

State **three** pieces of information that a data packet must contain.

[3]

7. [Maximum mark: 15]

23N.1.SL.TZ0.10

The staff at a doctor's practice consist of a receptionist and a doctor.

The patients' medical records and payments, the doctor's appointment calendar, and other important data are stored in a database on the central computer.

(a) Outline **one** security measure that can be taken to prevent unauthorized access to the patients' data stored on the central computer.

[2]

(b.i) Identify **one** cause of data loss. [1]

(b.ii) Describe **one** method that can be used to prevent data loss. [2]

A new vaccine has been distributed that would be of benefit to some of the doctor's patients. A large number of personalized letters need to be written to these patients, inviting them to visit the doctor's practice to be vaccinated.

(c) Describe how these letters could be automatically generated by a word processing application. [3]

When the doctor visits a patient in their home, she needs to be able to access the patient's medical records stored on the central computer in the practice.

(d) Outline **two** reasons for the use of a virtual private network (VPN) in this situation. [4]

A mobile data connection enables the doctor to access internet resources while visiting patients in their homes. Sometimes the data transfer speeds are slow.

(e) Explain why the speed of data transmission across a mobile network can vary. [3]

8. [Maximum mark: 15]

23M.1.SL.TZ2.11

Many organizations use a virtual private network (VPN) to enable employees working remotely to access files that are held on the organization's server.

(a) State **two** technologies that are required to provide a virtual private network (VPN). [2]

(b) Identify **two** factors that may affect the speed of data transmission. [2]

- (c) Explain why data compression would be used when data is transmitted. [3]

A large amount of sensitive data is stored online and needs to be protected.

- (d) Outline how encryption is used to protect data. [2]

- (e) Describe the role of a firewall. [2]

Employees are increasingly working from home.

- (f) Discuss the social impacts of this changed work pattern on employees. [4]

9. [Maximum mark: 4] 22N.1.SL.TZ0.5

A student posts images and videos on a public website of her friends at a party.

- (a) Outline **one** ethical issue with the student posting these images and videos. [2]

- (b) Outline **one** technical issue that may prevent the images and videos from being viewed. [2]

10. [Maximum mark: 15] 22N.1.SL.TZ0.9

A company has a local area network (LAN). Ethernet (a wired network) and WiFi (a wireless network) are the two ways to enable LAN connections.

The LAN is accessible to all employees through their personal accounts. At the office, employees can use either desktop computers for wired access to the LAN or personal laptops to connect wirelessly.

- (a) Identify **one** additional hardware component in a wireless LAN. [1]

(b) Distinguish between a wired network and a wireless network in terms of reliability of transmission. [4]

(c) Outline why a wireless network may be less secure than a wired network. [2]

Employees who are not in the office can access the company's resources over the internet using a virtual private network (VPN).

(d) Outline **two** features of a VPN that make it secure. [4]

(e) Packet switching is used for transmitting data.

Explain how data is transmitted by packet switching. [4]

11. [Maximum mark: 2] 22M.1.SL.TZ0.3
Identify **two** differences between a wide area network (WAN) and a local area network (LAN). [2]

12. [Maximum mark: 2] 22M.1.SL.TZ0.4
Outline the reason for compression when transmitting data. [2]

13. [Maximum mark: 2] 22M.1.SL.TZ0.10
Outline **one** reason for the use of standards in the construction of networks. [2]

14. [Maximum mark: 2] 21M.1.SL.TZ0.6
Identify **two** reasons why fibre optic cable would be preferred over wireless connectivity.

[2]

15. [Maximum mark: 1]

21M.1.SL.TZ0.9

Define the term *data packet*.

[1]